

Evaluating Usability Challenges in VR Games for Older Adults: A Comparison With and Without AI Assistance

Emily Kuang
Lassonde School of Engineering
York University
Toronto, Ontario, Canada
ekuang@yorku.ca

Abstract

Evaluating VR usability presents unique challenges, particularly when designing for older adults. This study explores how AI-generated suggestions might support UX evaluators in identifying usability problems in VR games. Eight evaluators reviewed videos of older adults interacting with a VR game under two conditions: with and without AI assistance. Older adults commonly encountered problems related to unclear instructions, confusing game mechanics, and poor menu navigation. Evaluators in the AI condition identified more usability issues overall, including additional categories such as font size and scoring confusion. While preliminary, these findings suggest that AI tools may help expand evaluators' perspectives, especially in contexts where they lack relevant lived experience.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Usability testing**.

Keywords

User experience; Usability analysis; Human-AI collaboration; Older adults; Virtual reality

ACM Reference Format:

Emily Kuang. 2025. Evaluating Usability Challenges in VR Games for Older Adults: A Comparison With and Without AI Assistance. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758343>

1 Introduction

Virtual reality (VR) offers immersive experiences and has shown promise as a form of physical activity for older adults [8]. However, they often face many usability challenges, which complicate navigation and task completion in VR [2, 13, 18]. While prior studies have explored older adults' verbalizations while using VR [4], they have not systematically analyzed the types of usability problems these users encounter. Separately, prior work has demonstrated the potential of AI to support usability analysis [3, 9, 10, 12, 17], but has not directly compared evaluator performance with and without AI

assistance. This study addresses these two gaps by (1) categorizing usability problems faced by older adults during VR gameplay, and (2) evaluating whether large language models (LLMs), specifically GPT-4, can assist UX evaluators in usability analysis.

We conducted a usability test with five older adults ($avg.age = 78.8, SD = 4.5$) who had limited or no prior VR experience. Participants played three VR games (e.g., blocking, volleyball, and skeet) while following a think-aloud protocol. We recorded their sessions and transcribed them for analysis. Next, we conducted a between-subjects study with eight UX evaluators: four analyzed the usability videos manually (baseline), and four used a custom GPT-4 assistant that suggested potential usability problems.

Our findings reveal ten distinct categories of usability problems, including unclear instructions, confusing game mechanics, and poor menu navigation. Evaluators using GPT-4 identified significantly more usability problems and uncovered two problem categories not found in the baseline condition. This study contributes categories of usability problems older adults face in VR gaming and preliminary evidence that AI support improves both the quantity of identified usability problems.

2 Generating Usability Problem Suggestions using GPT-4

We developed a custom conversational agent powered by GPT-4 to generate usability problem descriptions. Following OpenAI's GPT creation guide [14], we configured system instructions using the GPT Builder interface in ChatGPT. The custom GPT was given the following persona:

You are an experienced UX evaluator with extensive hands-on experience conducting user research and a deep understanding of UX methodologies [16]. You customize usability problem descriptions, ensure reliability in your analyses [5, 11], and actively question inconsistencies to clarify design goals [6].

We uploaded VR game screenshots, task descriptions, and video transcripts to the custom GPT, then prompted it to generate usability problem descriptions. Prior work suggests that generating multiple responses improves the breadth and consistency of ChatGPT's output [7, 15]. To increase coverage, we prompted the GPT three times per video and combined all unique suggestions. The custom GPT generated an average of 7 usability problems per video ($SD = 0.7$), for a total of 35 problems across five videos.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST Adjunct '25, Busan, Republic of Korea

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758343>

3 User Study

We recruited eight UX evaluators through professional networks and randomly assigned them to either the baseline or AI-assisted condition. Each group included four participants (three females, one male). Those in the baseline condition had an average age of 26.3 (SD=2.2) and 4.0 years of UX experience (SD=2.0), while those in the AI condition had an average age of 28.8 (SD=6.2) and 3.5 years of experience (SD=1.7). There were no significant differences in UX experience between the two groups.

All participants analyzed five usability videos across five sessions. In both conditions, they used a custom web-based tool with a video player and a problem panel for entering usability issues. The AI condition included an additional chatbox displaying GPT-4-generated usability suggestions. Following prior work showing that UX evaluators preferred AI suggestions timed immediately after relevant video segments [12], we adopted this approach. When an AI suggestion appeared, participants could *accept*, *edit*, or *ignore* it. This design encouraged critical engagement while allowing participants to maintain control over their final analyses.

We collected system logs, screen recordings, and all usability problems entered into the tool for subsequent analysis.

4 Results

4.1 Categories of Usability Problems that Older Adults Encountered in VR Games

Fig. 1 presents the categories of usability problems identified by participants in the baseline and AI conditions while evaluating VR games. Each participant reviewed five usability videos, and the number of identified problems varied across conditions.

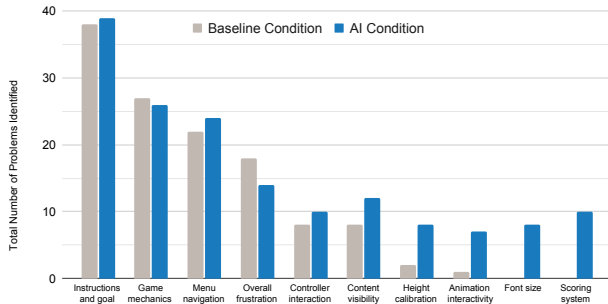


Figure 1: Bar chart showing the number of problems identified per category across the five usability videos in the baseline and AI conditions

The most frequent categories of usability problems were:

- (1) **Instructions and goals:** Older adults noted that unclear instructions led to confusion about game objectives. For example, in one game, some mistakenly believed they were supposed to hit the ball over a net instead of into a bin.
- (2) **Game mechanics:** Older adults struggled with button mappings and performing actions like pressing the trigger or reloading the skeet gun.

- (3) **Menu navigation and game selection:** The menu design was unclear. Older adults often did not realize they needed to use the joystick to navigate between games and press the 'A' button to make a selection.
- (4) **Overall frustration:** Some older adults expressed sentiments of frustration and loss of control, which resulted in disengagement from the gameplay experience.

Less frequently reported problems included: **controller interaction** (e.g., forgetting button locations), **content visibility** (e.g., missing objects in peripheral vision), **height calibration** (e.g., virtual objects positioned too high to reach comfortably), and **animation interactivity** (e.g., difficulty distinguishing decorative versus interactive elements).

Notably, evaluators in the AI condition identified two additional categories that were not mentioned by those in the baseline condition: **font size** (e.g., small text that was hard to read) and **scoring system** (e.g., confusion over the meaning of the onscreen numbers).

4.2 Number of Usability Problems Identified with and without AI Assistance

UX evaluators identified a total of 282 usability problems across all videos, with 124 problems in the baseline conditions and 158 in the AI condition. Fig. 2 shows a box plot of the number of problems identified per video, separated by condition. The box plots for the AI condition consistently appear above those for the baseline condition, indicating that AI assistance led participants to identify more usability problems. A two-sample t-test confirmed that this difference was statistically significant ($p < .01$), suggesting that AI-generated suggestions effectively supported evaluators in uncovering a greater number of usability problems.

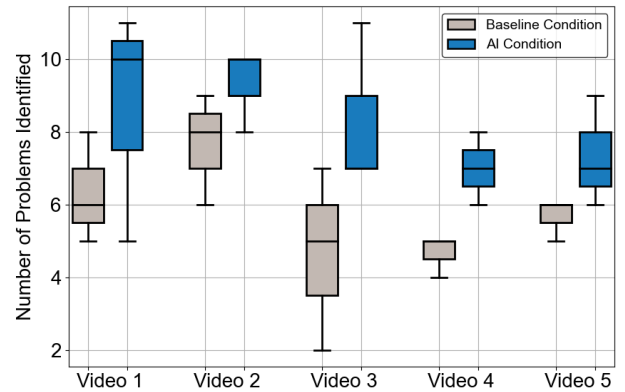


Figure 2: Box plot showing the number of problems identified per video in the baseline and AI conditions

5 Future Work

This study takes an initial step toward understanding how AI-generated suggestions can support usability evaluations of VR experiences for older adults. We plan to expand to other VR applications and recruit a larger, more diverse sample to identify

usability challenges across contexts. While AI assistance appears to increase the number of identified usability problems, further analysis is needed to evaluate their relevance and how AI shapes evaluators' experiences. We will investigate alternative ways to deliver AI feedback, such as on-demand suggestions, to better fit evaluators' workflows. We also aim to incorporate cognitive forcing functions, like timed reflection pauses, to encourage more critical engagement with AI feedback [1]. These approaches may improve evaluators' assessment of AI accuracy and enhance human-AI collaboration in usability analysis. Finally, we recognize the ethical risks of generating AI-driven insights without involving the target population. Future work will engage with older adults in designing and interpreting AI-supported evaluation tools.

References

- [1] Zana Bućinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1 (April 2021), 188:1–188:21. <https://doi.org/10.1145/3449287>
- [2] Qiuxin Du, Xiaoying Wei, Jiawei Li, Emily Kuang, Jie Hao, Dongdong Weng, and Mingming Fan. 2025. AI as a Bridge Across Ages: Exploring The Opportunities of Artificial Intelligence in Supporting Inter-Generational Communication in Virtual Reality. *Proc. ACM Hum.-Comput. Interact.* 9, 2 (May 2025), CSCW026:1–CSCW026:29. <https://doi.org/10.1145/3710924>
- [3] Peitong Duan, Jeremy Warner, Yang Li, and Bjoern Hartmann. 2024. Generating Automatic Feedback on UI Mockups with Large Language Models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 6, 20 pages. <https://doi.org/10.1145/3613904.3642782>
- [4] Mingming Fan, Qiwen Zhao, and Vinita Tibdewal. 2021. Older Adults' Think-Aloud Verbalizations and Speech Features for Identifying User Experience Problems. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 358, 13 pages. <https://doi.org/10.1145/3411764.3445680>
- [5] Asbjørn Følstad, Effie Lai-Chong Law, and Kasper Hornbæk. 2010. Analysis in Usability Evaluations: An Exploratory Study. In *Proceedings of the 6th Nordic Conference on Human-Computer Interaction: Extending Boundaries* (NordiCHI '10). Association for Computing Machinery, New York, NY, USA, 647–650. <https://doi.org/10.1145/1868914.1868995>
- [6] Eureka Foong, Darren Gergle, and Elizabeth M. Gerber. 2017. Novice and Expert Sensemaking of Crowdsourced Design Feedback. *Proc. ACM Hum.-Comput. Interact.* 1, CSCW (Dec. 2017), 45:1–45:18. <https://doi.org/10.1145/3134680>
- [7] Kostis Giannakopoulos, Argyro Kavadelis, Anas Aaqel Salim, Vassilis Stamatopoulos, and Eleftherios G. Kaklamanos. 2023. Evaluation of the Performance of Generative AI Large Language Models ChatGPT, Google Bard, and Microsoft Bing Chat in Supporting Evidence-Based Dentistry: Comparative Mixed Methods Study. *Journal of Medical Internet Research* 25, 1 (Dec. 2023), e51580. <https://doi.org/10.2196/51580> Company: Journal of Medical Internet Research Distributor: Journal of Medical Internet Research Institution: Journal of Medical Internet Research Label: Journal of Medical Internet Research Publisher: JMIR Publications Inc., Toronto, Canada.
- [8] Thwe Zar Chi Htut, Vimonwan Hiengkaew, Chutima Jalayondeja, and Mantana Vongsirinavarat. 2018. Effects of physical, virtual reality-based, and brain exercise on physical, cognition, and preference in older persons: a randomized controlled trial. *European Review of Aging and Physical Activity* 15, 1 (Oct. 2018), 1–12. <https://doi.org/10.1186/s11556-018-0199-5> Number: 1 Publisher: BioMed Central.
- [9] Emily Kuang. 2023. Crafting Human-AI Collaborative Analysis for User Experience Evaluation. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (CHIEA '23). Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/3544549.3577042>
- [10] Emily Kuang, Ehsan Jahangirzadeh Soure, Mingming Fan, Jian Zhao, and Kristen Shinohara. 2023. Collaboration with Conversational AI Assistants for UX Evaluation: Questions and How to Ask them (Voice vs. Text). In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI '23). Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3544548.3581247>
- [11] Emily Kuang, Xiaofu Jin, and Mingming Fan. 2022. "Merging Results Is No Easy Task": An International Survey Study of Collaborative Data Analysis Practices Among UX Practitioners. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3491102.3517647>
- [12] Emily Kuang, Minghao Li, Mingming Fan, and Kristen Shinohara. 2024. Enhancing UX Evaluation Through Collaboration with Conversational AI Assistants: Effects of Proactive Dialogue and Timing. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (CHI '24). Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3613904.3642168>
- [13] Ryan C. Moore, Jeffrey T. Hancock, and Jeremy N. Bailenson. 2023. From 65 to 103, Older Adults Experience Virtual Reality Differently Depending on Their Age: Evidence from a Large-Scale Field Study in Nursing Homes and Assisted Living Facilities. *Cyberpsychology, Behavior, and Social Networking* 26, 12 (Dec. 2023), 886–895. <https://doi.org/10.1089/cyber.2023.0188>
- [14] OpenAI. 2024. Creating a GPT | OpenAI Help Center. <https://help.openai.com/en/articles/8554397-creating-a-gpt>
- [15] Krishna Ronanki, Beatriz Cabrero-Daniel, and Christian Berger. 2024. ChatGPT as a Tool for User Story Quality Evaluation: Trustworthy Out of the Box?. In *Agile Processes in Software Engineering and Extreme Programming – Workshops (Lecture Notes in Business Information Processing)*, Philippe Kruchten and Peggy Gregory (Eds.). Springer Nature Switzerland, Cham, 173–181. https://doi.org/10.1007/978-3-031-48550-3_17
- [16] Sara L. Rynes, Marc O. Orlicky, and Robert D. Bretz Jr. 1997. Experienced Hiring Versus College Recruiting: Practices and Emerging Trends. *Personnel Psychology* 50, 2 (1997), 309–339. <https://doi.org/10.1111/j.1744-6570.1997.tb00910.x> <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1744-6570.1997.tb00910.x>
- [17] Ziyang Wang, Luyao Shen, Emily Kuang, Shumeng Zhang, and Mingming Fan. 2024. Exploring the Impact of Artificial Intelligence-Generated Content (AIGC) Tools on Social Dynamics in UX Collaboration. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (DIS '24). Association for Computing Machinery, New York, NY, USA, 1594–1606. <https://doi.org/10.1145/3643834.3660703>
- [18] Xiaoying Wei, Yizheng Gu, Emily Kuang, Xian Wang, Beiyan Cao, Xiaofu Jin, and Mingming Fan. 2023. Bridging the Generational Gap: Exploring How Virtual Reality Supports Remote Communication Between Grandparents and Grandchildren. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI '23). Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3544548.3581405>